WSPRadar.org reconstruction & comparison

SNR difference by time of day

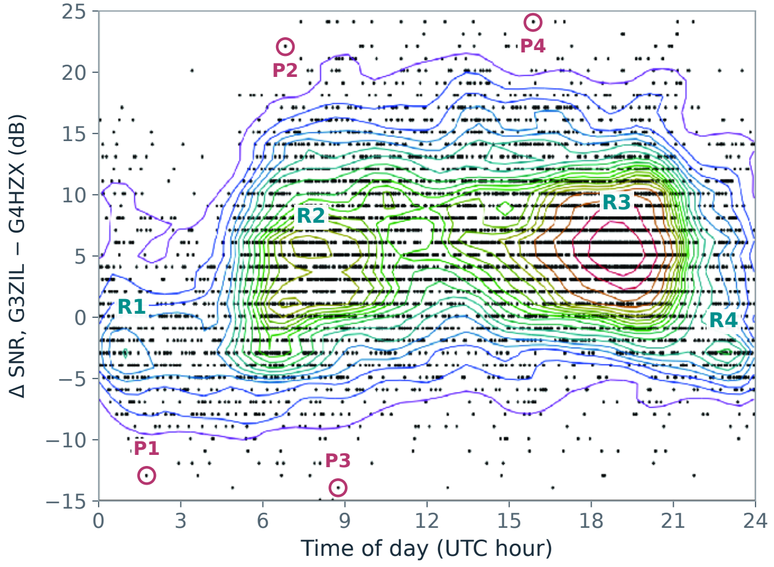
Referenced publication: Gwyn Griffiths (G3ZIL) and Nigel Squibb (G4HZX) (2017). *Improving HF Band SNR.*

Source figure: **Figure 6, p. 25** · Practical Wireless, October 2017, pp. 23-26

Demo: RX Benchmark | 5-7 April 2017 | 40 m | ΔSNR = Target G3ZIL (IO90HW) − Reference G4HZX (IO91) | B/C: 6 459 spot pairs | Reference correction 0.0 dB

A Original image from publication

Original scatter and contours · linear dB



R R1–R4: paper regions (A–C)

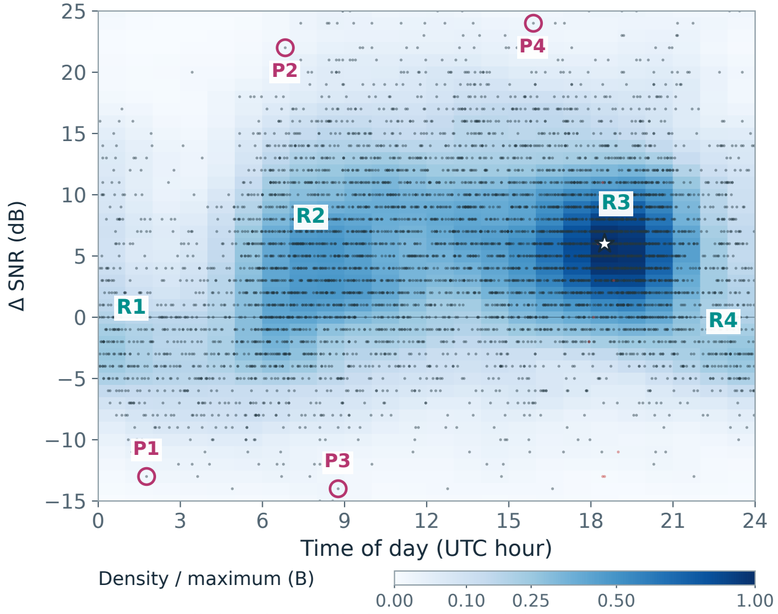
○ P1–P4: verification examples only (A–C)

Panel A: Original publication image, calibrated using its printed axes. R1–R4 locate the paper's regions.

P1–P4: Verification examples only, checking selected time-of-day and ΔSNR matches.

B Reconstruction

Smoothed observation density · linear dB



● Individual paired observations (B)

● Additional weaker-report combinations (B)

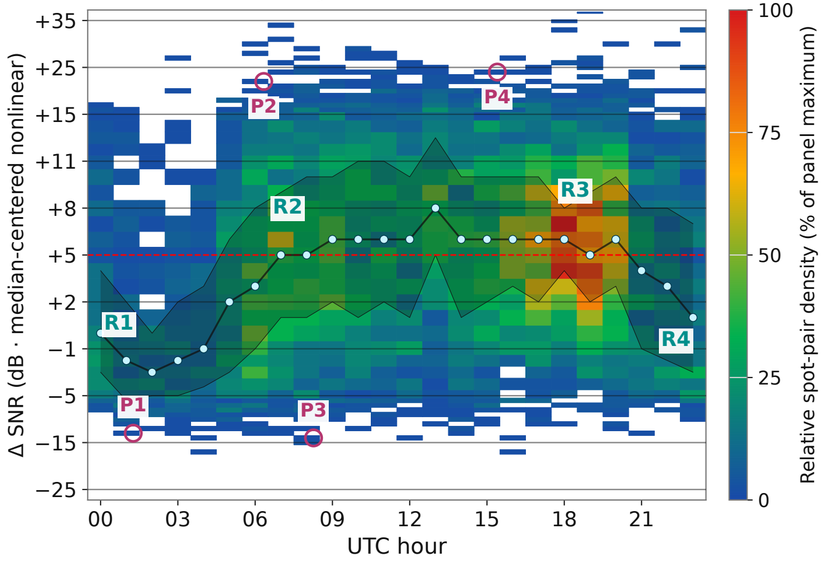
☆ Highest reconstructed density (B)

Panel B: Paired observations recalculated through WSPRadar. Dark dots show retained pairs; red dots show additional* weaker-report combinations, excluded from density. WSPRadar retains the strongest report from each receiver per cycle/path. Shading shows the retained pairs' density with fixed smoothing.

*see Figure 3 **Pairing and duplicate reports**

C WSPRadar view

Δ SNR by UTC Hour (1 h bins)



○ Median for each UTC hour (C)

▭ Middle 50% for each UTC hour (C)

Panel C: The same retained pairs in unsmoothed 1-hour × 1-dB cells, with hourly medians and the middle 50% (IQR). Read the native nonlinear dB axis carefully.

Original figure: p. 25, Figure 6. Selection: 5 April 00:00-7 April 23:45 UTC; distance <10,000 km.